Final 30K low-prevalence confirmation

30,000 frozen URLs • 5.00% phishing • unchanged models and thresholds

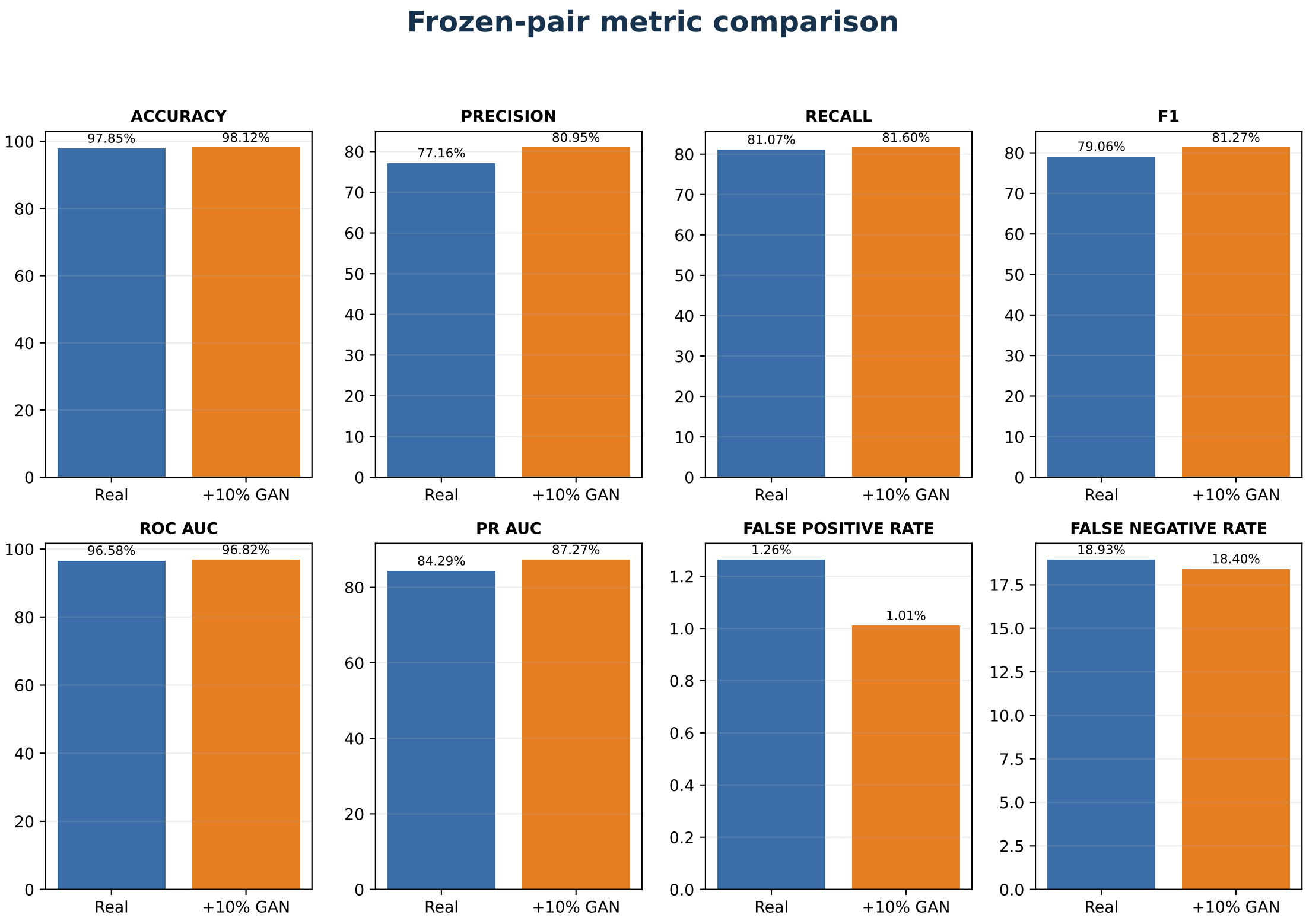
Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
HYB-MS real-only	97.85%	77.16%	81.07%	79.06%	96.58%	84.29%	1.26%	18.93%
HYB-MS +10% GAN	98.12%	80.95%	81.60%	81.27%	96.82%	87.27%	1.01%	18.40%

PRIMARY AUGMENTATION RULE PASSED

Recall, F1 and ROC-AUC improved without a higher FPR.

Paired exact McNemar p = 7.427e-05; GAN uniquely corrected 240 rows and uniquely worsened 160 rows.

No model, checkpoint, calibration value, threshold, or URL was changed after inference began.


Frozen-pair metric comparison
ACCURACY
97.85%
98.12%
PRECISION
77.16%
80.95%
RECALL
81.07%
81.60%
F1
79.06%
81.27%
ROC AUC
96.58%
96.82%
PR AUC
84.29%
87.27%
FALSE POSITIVE RATE
1.26%
1.01%
FALSE NEGATIVE RATE
18.93%
18.40%
Real
+10% GAN